

Victoria Nguyen

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EDUCATION

University of Michigan - Ann Arbor

Bachelor of Science in Computer Science, Minor in Entrepreneurship

Ann Arbor, MI

May 2028

- **Relevant Coursework:** Data Structure and C++, Discrete Structure, Entrepreneurship Practicum, Digital Product Design

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C++, HTML/CSS, Bash/CLI

Frameworks & Libraries: React, FastAPI, Node.js, Express, React Native, Streamlit, D3.js, Pandas, Plotly, Matplotlib, SDV

Technologies: MongoDB, REST APIs, Twilio API, Bluetooth LE, JSON, CSV, Git/GitHub, Jupyter Notebook

Specializations: Full-stack Web Development, AI/ML Applications, UI/UX Design

WORK EXPERIENCE

Michigan Medicine - Department of Computational Medicine & Bioinformatics

Research Assistant - Genomic AI Development

Ann Arbor, MI

Jan 2026 – Present

- Developing a conversational AI interface for genomic foundation models, enabling researchers to query DNA sequence predictions through natural language using React frontend and FastAPI backend architecture.
- Building an interactive genome browser component with real-time visualization of genomic regions, annotations, and model predictions using D3.js and custom rendering algorithms.
- Implementing RESTful API endpoints to bridge genomic foundation model outputs with user-facing application, handling complex bioinformatics data transformations and query processing.
- Collaborating with PhD researchers in Computational Biology to translate domain requirements into scalable web application features supporting genomic research workflows.

Michigan Medicine - Department of Pediatrics

Research Assistant - ABC Growing Kids Project

Ann Arbor, MI

2025 – Present

- Utilized Python and data visualization tools to analyze structured behavioral datasets, facilitating evidence-based development strategies for infant feeding behavior research.
- Conducted reliability testing on video data using Observer XT, achieving over 60% inter-rater reliability to ensure high-fidelity research outputs and consistent behavioral coding.

Michigan Medicine University Hospital

Nurse Aide II

Ann Arbor, MI

2025 – 2026

- Managed clinical data entry within EPIC EMR system on transplant unit; applied insights into standardized healthcare data requirements to inform structured data architectures in software projects.
- Collaborated within interdisciplinary care teams via EMR, identifying opportunities for improved system integration and communication protocols in high-acuity clinical settings.

EMU Center for Health Disparities

Research Assistant - Asian American Health

Ann Arbor, MI

2024 – 2025

- Performed quantitative analysis on survey data from 349 participants, uncovering key health disparities that informed strategic community health recommendations and local health initiatives.
- Designed technical communication materials to increase engagement for CMS Navigator programs, bridging complex health data with community outreach.

PROJECTS

Synthlab | Python, Streamlit, Plotly, SDV, GANs |

Dec 2025 – Present

- Built privacy-preserving synthetic data platform using Python and Streamlit with three generation methods (CTGAN, GaussianCopula, TVAE) for healthcare research validation and compliance testing.
- Implemented statistical quality assessment including KS testing, correlation analysis, and distribution comparison with interactive Plotly visualizations for data integrity validation.
- Designed privacy scoring system with exact-match detection and Distance-to-Closest-Record metrics to quantify re-identification risk in synthetic healthcare datasets.

Heartline Care Companion | React Native, Twilio, Python, Expo |

Oct 2025 – Nov 2025

- Built mobile care companion for transplant patients with mood, energy, and symptom check-ins modeled after Michigan Medicine clinical workflows and patient engagement protocols.
- Integrated Twilio SMS alerts notifying caregivers when symptom patterns exceed configurable thresholds, reducing emergency response time.
- Implemented structured vitals logging with JSON/CSV export and Bluetooth LE syncing for seamless clinician review and longitudinal data tracking.